



Data Communications & Networking

About Me

Ouldooz Baghban Karimi

Office: SRYE 5103

Web: www.cs.sfu.ca/~oba2/

Education

PhD, Computer Science

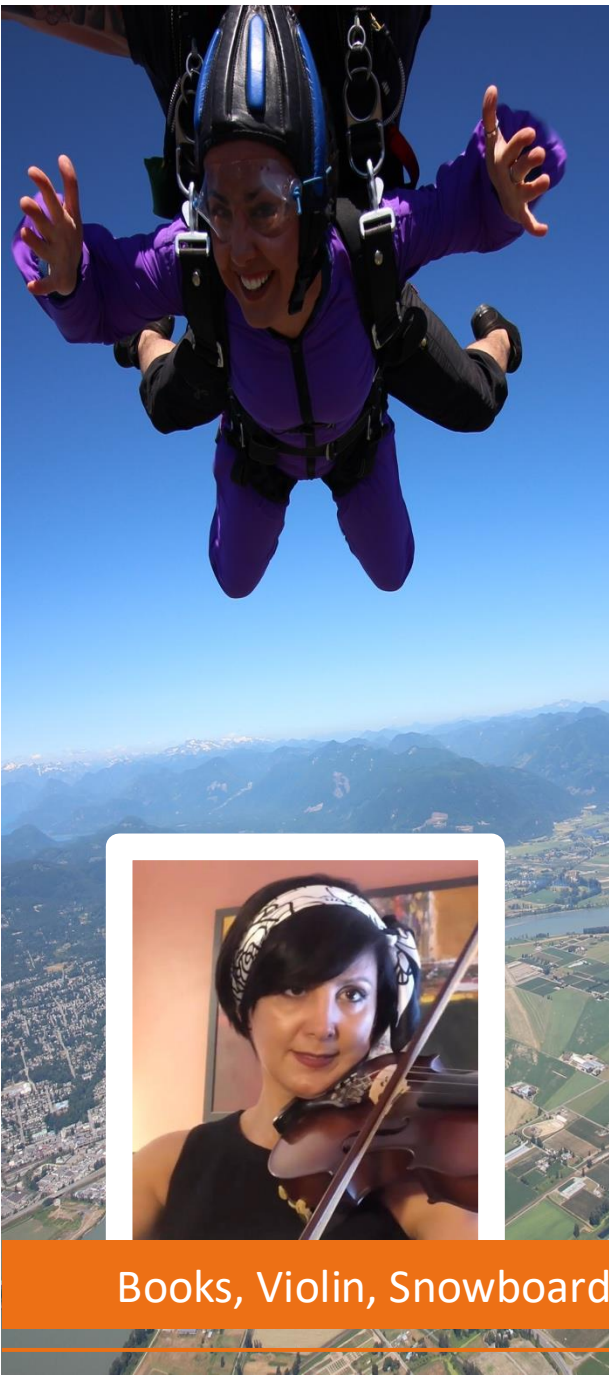
Interests

Networks and Systems

Human-Centric Networks

Diversity and Inclusion





Books, Violin, Snowboarding, Hiking, Travelling, Arts & Crafts

Ghazaleh Bakhtiariadzad (TA)

I recently completed my M.Sc. in Computer Science at Simon Fraser University, Network and Multimedia Systems Lab (NMSL), where my thesis focused on bandwidth optimization for cloud VR gaming using deep learning.

I have a background in distributed systems, cloud infrastructure, and applied ML.

I have served as a TA for the Data Communications & Networking course multiple times at SFU.



Ronghsheng Qian(TA)

Education:

SFU MSc Student

Research Interests:

Multimedia

Computer Vision

Compression

Denoising

Video Streaming

Website: <https://rsq0504.github.io/>



Mohsen Babanejad Khademloo (TA)

Education:

SFU MSc Cybersecurity

MSc in Computer Networks

System and Network Administrator

Several years of IT experience

Interests:

Infrastructure security

Threat mitigation



Topics Overview

- **Introduction to Networks:** Basic Principles, Architecture, and Reference Models
- **Application Layer:** Principles, Hypertext Transfer Protocol (HTTP), Mail, Domain Name System (DNS), Peer to Peer, Multimedia, Content Distribution Networks (CDN)
- **Transport Layer:** Reliable Transport, Connection, Congestion, Transport Control Protocol (TCP) & User Datagram Protocol (UDP)
- **Network Layer:** Data Plane, Control Plane, IP, Forwarding, Routing, Software Defined Networking (SDN)
- **Data Link Layer:** Concepts, Media Access, Local Area Networks (LANs)
- **Advanced Topics** (Time Permitting Throughout the Course)

Course Goals (1)

Gain knowledge about Computer Networks

At the end of this course, be able to **Understand** networking concepts, currently implemented and possible networking solutions including the concepts, components, and protocols in networking topics, including subjects in

- Network Overview
- Application Layer
- Transport Layer
- Network Layer
- Link Layer

Course Goals (2)

Gain analytical and design skills

At the end of this course, be able to **Apply** knowledge to **Analyze**, and **Evaluate** problems in the networking domain, including:

- Network Performance
- Transport Control Protocol Analysis (Window Size, Throughput, Fairness)
- Routing protocol analysis including calculating routing tables
- Link access performance calculation for covered protocols
- Identifying and designing required components for different types of communication
- Identifying security vulnerabilities of a given protocol or network design

Course Goals (3)

Gain hands-on skills

At the end of this course, be able to:

- **Evaluate** running and analysis of protocols
- **Create** Socket Programs
- **Create** (Configure) networks with specified addressing and subnetting

Course Components

- **Lectures:** Not recorded, Pre-recorded Youtube Videos + Active Learning (Not recorded)
Recordings are updated every Monday: On normal sessions for the last week. On OAS, for the upcoming week.
- **Interactive Sessions:** CourSys
- **Assignments:** CourSys
- One **Mini-Project** (Group Project: Groups of Two Students, File Submission to CourSys)
- **Four Quizzes:** In person, on paper
- **Midterm** (Writing Questions)
(90 mins, on paper, in person, Closed-book, one-page two-sided cheat sheet allowed)
- **Final** (Writing Questions)
(120 mins, on paper, in person, Closed-book, one-page two-sided cheat sheet allowed)

Communications

- Announcements
 - Class mailing list
- Active Learning Sessions
 - Assigned discussions
 - Group and class activities
 - Use Active Learning Sessions for additional lecture related questions and discussions
 - Questions and discussions only about the topic (Latest lecture)
- Office Hours (In-Person & Zoom)
 - Use office hours for longer questions

Communications

- Course mailing lists
 - ***cmpt-371-teaching@sfu.ca*** (Teaching Team)
 - ***cmpt-371@sfu.ca***: Announcements (registered automatically, rarely used)
- Email policy (cmpt-371-teaching)
 - Use email for private/administrative communication
 - Teaching Team Mailing list
 - [371 Urgent] Brief Description
 - [371 Office Appointment Request] Brief Description
 - [371 Request] Brief Description
 - [371 Question] Brief Description

Communications

- Discussion Board (CourSys)
 - Use discussion boards for
 - Quick questions and guidelines on lectures, homework, assignments, etc.
 - Engage with your classmates in discussions - Students
 - Find study or project groups - Students
- Respectful and Understanding
 - Polite
 - Direct
 - Timely

Office Hours

- **Ouldooz:** Wednesdays, 4-5pm, In-Person, **SRYE 5103** (By Appointment)

Office: **SRYE 5103**

Starting the second week of the semester

Book appointments on Canvas Calendar

- **Ghazaleh, Rongsheng, and Mohsen:** Drop-in online sessions

Time & Zoom information on Canvas

Starting the second week of the semester

For the By Appointment Office Hours, please book your appointments 24 hours before the session. Please do not book more than two appointments in one day.

Grading

- Assignments & Mini Project **(30%)**
 - Four Assignments (15%)
 - One Mini Project (15%)
 - Late policy (all items): -25% each 24 hours (or less)
- Interactive Sessions & Quizzes **(15%)**
 - Interactive Sessions (5%)
 - Quizzes (10%)
 - Late policy: No late submission accepted
- Midterm & Final **(55%)**
 - Midterm: 20%
 - Final: 35%
 - Late policy: No late submission accepted

Overall passing grade on the **weighted average of exams (midterm & final)** to obtain a clear pass (C- or better).

Letter Grades

Overall passing grade (50%) on the **weighted average of exams (midterm & final)** to obtain a clear pass (C- or better).

- If the student does not meet the clear pass requirement, but overall grade is >50%, the course grade will be D.
- If the student meets the clear pass requirement, the course grade will be calculated using the overall grade percentage (not only the exams) and the table below:

Letter Grade	From	To
A+	95	100
A	90	94.99
A-	85	89.99
B+	80	84.99
B	75	79.99
B-	70	74.99
C+	65	69.99
C	60	64.99
C-	55	59.99
D	50	54.99
F	0	49.99

Course Resources

Course Web Page: <https://ouldoozbk.github.io/teaching/summer2026/DCN/>

Acronyms used in the schedule:

- **AL:** Active Learning (In-person Problem Solving Session)
- **VL:** Video Lecture (YouTube Topic-based Videos)
- **IS:** Interactive Session (Online Canvas Repeatable Quiz + Hands on Wireshark & Programming)
- **OAS:** Online-only Asynchronous Session (Assigned online videos and activities)
- **SR:** Strongly Recommended Reading
- **OR:** Optional but Recommended Reading
- **OP:** Optional Reading

Textbook

Textbook and (edited) Slides:

Computer Networking: A Top-Down Approach: http://gaia.cs.umass.edu/kurose_ross/
James Kurose, Keith Ross, 8th Edition, Pearson (2020)

Reference:

Computer Networks: A Systems Approach: <https://www.systemsapproach.org/book.html>

New Book:

The Real Internet Architecture: <https://cna.scholar.princeton.edu/>
Pamela Zave, Jennifer Rexford, First Edition, Princeton University Press (2024)